

Solving the Heterogeneous-Agent Model

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2 key relationships.

$$AS: \quad y = y^S(x) = f(x) \quad k$$

$$AD: \quad y = \sigma(x) \times [y^S(x) + \mu/p]$$

Need to solve a system of 2 equations:

$$\begin{cases} y = y^S(x) \\ y = \sigma(x) [y^S(x) + \mu/p] \end{cases}$$

$$\Rightarrow \begin{cases} y = y^S(x) \\ y = \sigma(x) [\underline{y} + \mu/p] \end{cases}$$

(substitution)

$$\Rightarrow \begin{cases} y = y^S(x) \\ y = \frac{\sigma(x)}{1 - \sigma(x)} (\mu/p) \end{cases}$$

$$\left| \frac{\sigma(x)}{1 - \sigma(x)} = X^\varepsilon [1 + \tau(x)]^{1-\varepsilon} \quad \text{b/c} \quad \sigma(x) = \frac{X^\varepsilon [1 + \tau(x)]^{1-\varepsilon}}{1 + X^\varepsilon [1 + \tau(x)]^{1-\varepsilon}} \right|$$

$$\text{so } 1 - \sigma(x) = \frac{1}{1 + X^\varepsilon [1 + \tau(x)]^{1-\varepsilon}}$$

The system that describes the model therefore is

$$\begin{cases} y = y^S(x) \quad \leftarrow \text{increasing in } x \\ y = \frac{x^\varepsilon}{[1 + \tau(x)]^{\varepsilon-1}} \left(\frac{p}{P} \right) \quad \leftarrow \text{if } p \text{ is fixed, decreasing in } x \end{cases}$$

Define the aggregate demand curve

$$y^d(x, p) = \frac{x^\varepsilon}{[1 + \tau(x)]^{\varepsilon-1}} \cdot \frac{p}{P}$$

then the model is given by the following system:

$$\begin{cases} y = y^S(x) \\ y = y^d(x, p) \end{cases}$$

Market tightness is implicitly given by:

$$y^S(x) = y^d(x, p)$$

As in representative-agent model:

- Tightness equalizes AD & AS curves
- AD & AS curves have same expression, same properties

Once tightness is obtained

can compute aggregate variables:

- $p = p^n(x)$
- $y = y^s(x)$
- $c = y / [1 + \tau(x)]$
- $m = \mu$ (Walras's Law)
- $v = y / q(x)$

$y_i, c_i, m_i, v_i \rightarrow$ can be computed from μ_i, k_i and tightness x for all i

- $y_i = \sigma(x) [f(x, k_i + \mu_i/p)]$
- $c_i = y_i / [1 + \tau(x)]$
- $\frac{m_i}{p} = [1 - \sigma(x)] [f(x) k_i + \mu_i/p]$
- $v_i = y_i / q(x)$